

A Certified and Differentiable Parametric Solver for Binary-Black-Hole Puncture Initial Data

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Constraint-satisfying initial data for binary black holes require the repeated solution of a non-linear elliptic boundary-value problem throughout the physical parameter space. We present a parametric initial-data solver that combines sparse-grid spectral interpolation with Newton refinement to construct certified, differentiable initial data over a chosen family of binary configurations. Interpolation is used only to provide warm starts for the underlying elliptic solver, rather than to replace it. A small number of Newton iterations refines each interpolated solution to a constraint residual $\|R\|_\infty \leq 10^{-10}$, independent of the interpolation error, thereby certifying every generated initial-data set. A JAX implementation further provides derivatives of the certified initial data with respect to the physical parameters, enabling gradient-based parameter targeting, optimization, and sensitivity analysis. We demonstrate the method for quasi-circular binary-black-hole puncture initial data by constructing four-dimensional aligned-spin and eight-dimensional general-spin models. The resulting initial data are validated against TwoPunctures, and the method is applied to certified targeting of the physical ADM mass and angular momentum and to effective-potential eccentricity reduction using a dedicated (b, P_t) model.

I. INTRODUCTION

Every numerical-relativity simulation of a binary black hole begins by solving the Einstein constraint equations on the initial time slice. For conformally flat, maximally sliced puncture data, this is a nonlinear elliptic boundary-value problem: the Hamiltonian constraint for the conformal factor in Lichnerowicz–York form [1–4]. This problem must be solved once for each configuration.

Modern applications, however, rarely require a single solution. Parameter surveys, waveform catalogs, eccentricity reduction, and mass and spin targeting all require the same elliptic problem to be solved repeatedly across the binary-black-hole parameter space. The repeated solution of the constraint equations therefore becomes the dominant recurring computational cost [5]. The computational challenge is therefore not solving a single elliptic problem, but efficiently solving a family of closely related elliptic problems across the physical parameter space.

Reduced-order modeling is a mature response to exactly this kind of repeated computation. In gravitational-wave physics, however, it has been developed almost exclusively for *waveforms*: reduced-basis template banks and surrogate models reproduce waveforms across parameter space at a tiny fraction of the cost [6–9]. The elliptic initial-data solve that precedes every numerical-relativity evolution has largely remained outside this development. Reduced-basis emulation does exist for the neutron-star sector, for example through Tolman–Oppenheimer–Volkoff emulators over equation-of-state

parameters [10], but no comparable parametric solver exists for binary-black-hole puncture initial data.

From the perspective of numerical analysis, this gap is somewhat surprising. Parametric collocation, sparse interpolation, and reduced-basis methods for parameter-dependent elliptic partial differential equations are well-established [11–15]. When the parameter-to-solution map is analytic, the interpolation error decays exponentially with the number of parameter samples at a rate determined by the nearest singularity in the complex parameter plane through the Bernstein-ellipse estimate [16]. These results suggest that binary-black-hole puncture initial data should also admit efficient parametric representations over suitable regions of parameter space.

Several recent developments have addressed complementary aspects of this problem. Mendes *et al.* [5] accelerate parameter targeting through a Broyden iteration that repeatedly re-solves the constraint equations. Tommasini *et al.* [17] use Gaussian-process regression trained on the SXS catalogue to predict low-eccentricity orbital parameters, reducing the number of trial evolutions required for eccentricity reduction. Zhou *et al.* [18] introduce a physics-informed neural-network solver for the puncture Hamiltonian constraint on a single configuration and identify parameterized initial-data solvers as an important direction for future work. Analytic perturbative constructions [19] and neural representations of spacetime metrics [20] provide additional complementary approaches. None of these methods, however, constructs a parametric solver that can certify the constraint satisfaction, and provide exact derivatives of the resulting initial data with respect to the physical parameters.

This paper addresses this gap by introducing a parametric solver for binary-black-hole puncture initial data

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with certified constraint satisfaction and exact differentiability with respect to the physical parameters. Rather than replacing the elliptic solve with an approximation, the method constructs a sparse interpolant that provides a high-quality initial guess for the nonlinear solve, after which a small number of Newton iterations recover a fully converged solution satisfying the Einstein constraint equations. We demonstrate the approach for both a four-dimensional aligned-spin model and an eight-dimensional general-spin quasi-circular model, and apply it to two representative tasks: targeting physical parameters and reducing orbital eccentricity.

The proposed solver combines three properties that, to our knowledge, have not previously been demonstrated together for binary-black-hole puncture initial data: it is parametric, provides certified constraint satisfaction, and is exactly differentiable with respect to the physical parameters. First, it provides *certified constraint satisfaction*. Every queried parameter is refined by Newton iterations until the constraint residual satisfies $\|R\|_\infty \leq 10^{-10}$, ensuring that every returned initial datum satisfies the Einstein constraints to a prescribed tolerance. The sparse interpolant therefore accelerates the computation but does not replace the underlying elliptic solver. Second, it provides *exact differentiability*. The entire construction is implemented in JAX [21], making the resulting parameter-to-solution map differentiable with respect to the physical parameters. These derivatives are obtained automatically through algorithmic differentiation.

The remainder of this paper is organized as follows. Section II reviews the puncture initial-data formulation and the resulting nonlinear elliptic constraint equation. Section III describes the spectral elliptic solver that produces a single constraint-satisfying datum. Section IV builds the parametric solver over it, combining sparse interpolation, Newton refinement that certifies every returned datum, and a differentiable JAX implementation. Section V constructs and characterizes the four-dimensional aligned-spin and eight-dimensional general-spin quasi-circular models, including their convergence and the analyticity limits of the parameter map. Section VI applies the method to certified parameter targeting and effective-potential eccentricity reduction. Finally, Section VII discusses the limitations, implications, and conclusions of this work. Validation of the initial data against TwoPunctures is collected in Appendix A.

II. PUNCTURE INITIAL DATA

We briefly review the puncture formulation of binary-black-hole initial data used throughout this work. We adopt conformally flat, maximally sliced initial data within York’s conformal transverse–traceless decomposition [1, 4], rather than the conformal thin-sandwich formulation [22]. Under these assumptions, the momentum constraint is satisfied analytically, leaving a single nonlinear elliptic equation for the conformal factor.

A. Constraint and puncture decomposition

In the conformal decomposition, the spatial metric and extrinsic curvature are $\gamma_{ij} = \psi^4 f_{ij}$ and $K^{ij} = \psi^{-10} \hat{A}^{ij}$, where f_{ij} is the flat conformal metric and $\hat{A}^{ij}$ is the conformal (Bowen–York) extrinsic curvature. We assume maximal slicing, $K \equiv f_{ij} K^{ij} = 0$. If $\hat{A}^{ij}$ is transverse with respect to the flat metric, $\partial_i \hat{A}^{ij} = 0$, the momentum constraint is satisfied analytically (Sec. II B), while the Hamiltonian constraint reduces to the nonlinear Lichnerowicz–York equation

$$\Delta\psi = -\frac{1}{8}\psi^{-7}\hat{A}_{ij}\hat{A}^{ij}, \quad (1)$$

where Δ is the flat Laplacian and indices are raised and lowered with f_{ij} . This is the only equation that must be solved numerically.

Equation (1) is defined on an unbounded domain and is singular at each puncture because $\hat{A}^2$ diverges there. Following Brandt and Brüggmann [3], we decompose the conformal factor as $\psi = \psi_{\text{BL}} + u$, where the N -puncture Brill–Lindquist background is

$$\psi_{\text{BL}} = 1 + \sum_X \frac{m_X}{2r_X}, \quad (2)$$

where $r_X \equiv |\mathbf{x} - \mathbf{x}_X|$ is the coordinate distance to puncture X . The Brill–Lindquist term contains all $1/r_X$ singularities and is harmonic away from the punctures. The regular correction therefore satisfies

$$\Delta u = -\frac{1}{8}(\psi_{\text{BL}} + u)^{-7} \hat{A}_{ij} \hat{A}^{ij}, \quad (3)$$

where, for the binary, we take $N = 2$ punctures of bare masses m_A and m_B located at $z = \pm b$. Solving Eq. (3) for u determines the complete initial data.

B. Bowen–York source term

The momentum constraint is satisfied analytically by the Bowen–York extrinsic curvature [2],

$$\hat{A}^{ij} = \sum_{X \in \{A, B\}} (\hat{A}_{P, X}^{ij} + \hat{A}_{S, X}^{ij}), \quad (4)$$

where the momentum and spin contributions are

$$\hat{A}_{P, X}^{ij} \equiv \frac{3}{2r_X^2} \left[P_X^i n_X^j + P_X^j n_X^i - (f^{ij} - n_X^i n_X^j)(\mathbf{P}_X \cdot \mathbf{n}_X) \right], \quad (5)$$

$$\hat{A}_{S, X}^{ij} \equiv \frac{3}{r_X^3} (v_X^i n_X^j - v_X^j n_X^i), \quad (6)$$

where $\mathbf{P}_X$ and $\mathbf{S}^X$ are the linear momentum and spin of puncture X , respectively, $\mathbf{v}_X \equiv \mathbf{S}^X \times \mathbf{n}_X$, and $\mathbf{n}_X \equiv (\mathbf{x} - \mathbf{x}_X)/r_X$ is the unit radial vector centered on puncture

X . The tensor f^{ij} in Eq. (5) is the inverse flat conformal metric; in the Cartesian coordinates used to construct the tensor, $f^{ij} = \delta^{ij}$. Each Bowen–York contribution is transverse and trace-free with respect to f_{ij} . Their sum therefore satisfies the momentum constraint analytically for arbitrary momenta, spins, and separations.

The Hamiltonian constraint (3) depends only on the scalar contraction $\hat{A}_{ij}\hat{A}^{ij}$. Defining $\hat{A}_X \equiv \hat{A}_{P,X} + \hat{A}_{S,X}$ as the total tensor for puncture X , the contraction decomposes into two self-contractions and a cross term,

$$\hat{A}_{ij}\hat{A}^{ij} = \hat{A}_A:\hat{A}_A + 2\hat{A}_A:\hat{A}_B + \hat{A}_B:\hat{A}_B. \quad (7)$$

Each self-contraction is

$$\begin{aligned} \hat{A}_X:\hat{A}_X &= \frac{9}{2r_X^4} [P_X^2 + 2(\mathbf{P}_X \cdot \mathbf{n}_X)^2] \\ &\quad + \frac{18}{r_X^6} |\mathbf{S}^X \times \mathbf{n}_X|^2 + \frac{18}{r_X^5} (\mathbf{P}_X \times \mathbf{S}^X) \cdot \mathbf{n}_X, \end{aligned} \quad (8)$$

where the three terms are the momentum, spin, and momentum–spin contributions, respectively. The cross term is

$$\begin{aligned} 2\hat{A}_A:\hat{A}_B &= \frac{9}{2r_A^2 r_B^2} \Pi_{AB} + \frac{36}{r_A^3 r_B^3} \Sigma_{AB} \\ &\quad + \frac{18}{r_A^2 r_B^3} \Xi_{AB} + \frac{18}{r_A^3 r_B^2} \Xi_{BA}, \end{aligned} \quad (9)$$

where the momentum–momentum (Π), spin–spin (Σ), and momentum–spin (Ξ) invariants are

$$\begin{aligned} \Pi_{AB} &\equiv 2(\mathbf{P}_A \cdot \mathbf{n}_B)(\mathbf{P}_B \cdot \mathbf{n}_A) - 3(\mathbf{P}_A \cdot \mathbf{n}_A)(\mathbf{P}_B \cdot \mathbf{n}_B) \\ &\quad + 2(\mathbf{n}_A \cdot \mathbf{n}_B)[(\mathbf{P}_A \cdot \mathbf{P}_B) \\ &\quad \quad + (\mathbf{P}_A \cdot \mathbf{n}_A)(\mathbf{P}_B \cdot \mathbf{n}_A) \\ &\quad \quad + (\mathbf{P}_B \cdot \mathbf{n}_B)(\mathbf{P}_A \cdot \mathbf{n}_B)] \\ &\quad + (\mathbf{P}_A \cdot \mathbf{n}_A)(\mathbf{P}_B \cdot \mathbf{n}_B)(\mathbf{n}_A \cdot \mathbf{n}_B)^2, \end{aligned} \quad (10)$$

$$\Sigma_{AB} \equiv (\mathbf{v}_A \cdot \mathbf{v}_B)(\mathbf{n}_A \cdot \mathbf{n}_B) + (\mathbf{v}_A \cdot \mathbf{n}_B)(\mathbf{v}_B \cdot \mathbf{n}_A), \quad (11)$$

$$\begin{aligned} \Xi_{AB} &\equiv (\mathbf{P}_A \cdot \mathbf{v}_B)(\mathbf{n}_A \cdot \mathbf{n}_B) + (\mathbf{P}_A \cdot \mathbf{n}_B)(\mathbf{n}_A \cdot \mathbf{v}_B) \\ &\quad + (\mathbf{P}_A \cdot \mathbf{n}_A)(\mathbf{n}_A \cdot \mathbf{n}_B)(\mathbf{n}_A \cdot \mathbf{v}_B), \end{aligned} \quad (12)$$

and where Ξ_{BA} follows from exchanging A and B .

C. Regularity of the puncture formulation

The puncture decomposition isolates the singular part of the conformal factor in the analytic Brill–Lindquist background ψ_{BL} . Near puncture X , $\psi_{\text{BL}}^{-7} \sim r_X^7$, whereas the momentum and spin self-contractions scale as $\hat{A}_{P,X}^2 \sim r_X^{-4}$ and $\hat{A}_{S,X}^2 \sim r_X^{-6}$, respectively. Hence, their contributions to the effective source in Eq. (3) vanish at least linearly as $r_X \rightarrow 0$. The cross terms are likewise regular and vanish at the puncture. Consequently, u is smooth throughout the compactified computational domain, making it well suited to the spectral discretization introduced in Sec. III.

III. CONSTRAINT SOLVER

We solve Eq. (3) for the regular correction u by spectral collocation on a single spectral patch. We use the Ansorg–Brügmann–Tichy (TwoPunctures) spectral method [23], extended with a Fourier expansion in ϕ for non-axisymmetric configurations. This solver provides the reference solution on which the parametric method of Sec. IV is built and against which it is validated.

A. The prolate-spheroidal single patch

The Ansorg–Brügmann–Tichy (ABT) coordinate system [23] maps the two-puncture domain onto a single compact spectral patch. Starting from cylindrical coordinates (ρ, z, ϕ) , it introduces prolate-spheroidal coordinates whose foci coincide with the punctures at $z = \pm b$, together with a compactified radial coordinate. With $A \in [0, 1]$ and $B \in [-1, 1]$, the transformation to cylindrical coordinates is

$$\frac{\rho}{b} = \frac{2A\sqrt{1-B^2}}{1-A^2}, \quad \frac{z}{b} = \frac{(1+A^2)B}{1-A^2}. \quad (13)$$

The coordinate edges have a clean meaning: $A = 1$ is spatial infinity, $A = 0$ is the inner axis segment $|z| \leq b$, the edges $B = \pm 1$ are the outer axis $|z| \geq b$, and the two corners ($A = 0, B = \pm 1$) are the punctures themselves.

Because the ABT coordinates are orthogonal, the flat-space Laplacian takes the separable form

$$\begin{aligned} \Delta u &= \frac{(1-A^2)^2}{b^2 \mathcal{D}} \left[\frac{(1-A^2)^2}{4A} \partial_A(A \partial_A u) \right. \\ &\quad \left. + \partial_B[(1-B^2) \partial_B u] \right] + \frac{\partial_\phi^2 u}{\rho^2}, \end{aligned} \quad (14)$$

with $\mathcal{D} \equiv (1+A^2)^2 - B^2(1-A^2)^2$ and ρ from Eq. (13).

In a Fourier basis in ϕ , the azimuthal derivative diagonalizes, $\partial_\phi^2 \rightarrow -m^2$, so each Fourier mode can be treated independently. The linear Lichnerowicz operator therefore decomposes into independent two-dimensional blocks, eliminating the need to assemble the full three-dimensional dense matrix.

The regular correction is represented nodally: the unknown is the tensor of collocation values $U_{ijk} \equiv u(A_i, B_j, \phi_k)$, rather than the modal coefficients (c_{nl}^m below). This representation is equivalent to the truncated spectral expansion

$$u(A, B, \phi) = \sum_{|m| \leq N_\phi/2} \sum_{n=0}^{N_A} \sum_{l=0}^{N_B-1} c_{nl}^m \phi_{nl}^m(A, B, \phi) \quad (15)$$

with basis functions,

$$\phi_{nl}^m(A, B, \phi) \equiv T_n(2A-1) P_l(B) (1-B^2)^{|m|/2} e^{im\phi}. \quad (16)$$

The basis consists of a Fourier expansion in the azimuthal angle ϕ , (with $c_{nl}^{-m} = (c_{nl}^m)^*$ since u is real), Chebyshev polynomials T_n in the compactified radial coordinate A , and Legendre polynomials P_l in the angle B . The associated-Legendre prefactor $(1 - B^2)^{|m|/2}$ carries the exact behaviour of the regular m -mode at the outer axis $B = \pm 1$. For odd m this yields a square-root branch point that a bare polynomial can only resolve algebraically, so we solve for the smooth polynomial factor and differentiate the prefactor analytically, restoring spectral convergence. The coefficients c_{nl}^m are not formed explicitly. The azimuthal FFT maps the N_ϕ node values to the modes m , and within each meridian block the A - and B -derivatives act on the nodal values through the Chebyshev and Legendre differentiation matrices.

A solved field must also be evaluated *off* the collocation grid at arbitrary coordinates (A, B, ϕ) . Although this could be done by forming the coefficients c_{nl}^m and evaluating the expansion in Eq. (15), we instead interpolate the nodal values directly. We use the numerically stable barycentric-Lagrange formula [16] in A and B , applied to the smooth polynomial factor with the analytic $(1 - B^2)^{|m|/2}$ prefactor restored for each mode, and equispaced trigonometric interpolation in ϕ . This procedure is not an approximation to the truncated spectral representation: the barycentric interpolant is the unique polynomial of the retained degree passing through the collocation values, while the trigonometric interpolant is the corresponding unique band-limited periodic interpolant. Consequently, the interpolated field is mathematically identical to the truncated expansion of Eqs. (15) and (16), evaluated in a coefficient-free form.

We place Chebyshev–Gauss–Lobatto nodes in A (N_A+1 nodes, with $A=1$ at spatial infinity and $A=0$ on the inner axis) and Gauss–Legendre nodes in B (N_B interior nodes that avoid the polar edges $B = \pm 1$ and, being a full nodal grid, carry both z -parities). Boundary conditions are imposed by row-replacement: Dirichlet $u = 0$ at the $A = 1$ (infinity) edge and Neumann $\partial_A u = 0$ on the $A = 0$ inner axis; on that inner axis only the $m = 0$ mode is regular, so we impose $u_m = 0$ there for $m \neq 0$, consistent with the $m = 0$ Neumann row. The Laplacian’s rows span a huge dynamic range: the $(1 - A^2)^4$ factor is tiny near infinity and the $1/A$ factor large near the inner axis, giving a raw condition number $\sim 10^{13}$ – 10^{15} . Using row equilibration, i.e. scaling each row by $1/\max_j |M_{ij}|$, an exact rescaling of the equations that leaves the solution unchanged, collapses the condition number to $\sim 10^4$ and takes the solve to machine precision.

B. Newton iteration with an analytic Jacobian

We solve Eq. (3) for the nodal field $U_{ijk} = u(A_i, B_j, \phi_k)$ using Newton iterations. Starting from an initial iterate $u^{(0)}$, each step solves the linearized system

$$J(u^{(k)}) \delta u^{(k)} = -R(u^{(k)}), \quad (17)$$

for the increment $\delta u^{(k)}$ and updates $u^{(k+1)} = u^{(k)} + \delta u^{(k)}$. The residual and its exact Jacobian $\partial R/\partial U$ are

$$R(u) \equiv M_0 u + \frac{1}{8} (\psi_{BL} + u)^{-7} \hat{A}^2|_{\text{int}}, \quad (18)$$

$$J(u) \equiv M_0 - \frac{7}{8} \text{diag}[(\psi_{BL} + u)^{-8} \hat{A}^2]_{\text{int}}, \quad (19)$$

where M_0 denotes the discretized Laplacian of Eq. (14), including the row-replaced boundary conditions, and the subscript “int” indicates that the nonlinear source is applied only on the interior rows. Because the Bowen–York source couples the Fourier modes whenever the data are non-axisymmetric, we evaluate the nonlinear term pseudo-spectrally on the ϕ collocation grid.

The structure of the Jacobian dictates the organization of the solver. The linear operator M_0 is block-diagonal in the azimuthal Fourier mode m ($\partial_\phi^2 \rightarrow -m^2$), so each mode reduces to an independent two-dimensional (A, B) solve that is factored once by its row-equilibrated LU decomposition. The nonlinear contribution breaks this decoupling. Its Jacobian,

$$D_{\text{nl}}(A, B, \phi) \equiv -\frac{7}{8} (\psi_{BL} + u)^{-8} \hat{A}^2, \quad (20)$$

depends on ϕ for non-axisymmetric data, and multiplication by a ϕ -dependent function corresponds to convolution in Fourier space. Consequently, the nonlinear term couples the azimuthal modes, making the full Jacobian dense across m .

We therefore solve the coupled Newton system with the full Jacobian Eq. (19) in matrix-free form, requiring only the matrix–vector product $J \delta u$. It is assembled exactly as the residual: the linear per-mode blocks M_0 plus the nonlinear coupling computed as $\text{rfft}_\phi[D_{\text{nl}} \delta u]$, i.e. a pointwise multiplication in ϕ followed by an FFT. The resulting linear system Eq. (17) is solved with GMRES, preconditioned by the block-diagonal linear operator. Since this preconditioner differs from the full Jacobian only through the omission of the Fourier-mode coupling, a few GMRES iterations suffice, while the outer Newton iteration retains quadratic convergence until the row-equilibrated residual stagnates, at which point the best iterate is returned. This Newton–Krylov solve is also the operation invoked by the certified polish of Sec. IV C.

Finally, we specify the norm used to monitor convergence. Because the prolate operator’s rows span the wide dynamic range of Sec. III A, the *raw* nodal residual $\|R\|_\infty$ is dominated by the badly-scaled rows near the inner axis: it floors well above machine precision and even *rises* with resolution. This is a scaling artifact, not a failure to satisfy the constraint. Row equilibration removes it: scaling each row by $s_i^{(m)} \equiv \max_j |(M_0^{(m)})_{ij}|$ leaves the solution unchanged, so the honest measure is the *row-equilibrated* residual $\max_m \max_i |\hat{R}_i^{(m)} / s_i^{(m)}|$, and it is this well-conditioned norm that the Newton iteration drives to zero. This residual is nonetheless a *solver-internal* diagnostic; the *physical* constraint violation of the emitted data, computed by a generic finite-difference monitor on an evolution grid, with no reference to this norm, is verified independently in Appendix A 3.

IV. PARAMETRIC SOLVER

So far we have described how to obtain one constraint-satisfying datum. The contribution of this paper is to obtain the datum *as a function of* the physical parameters, without a full elliptic solve at query time.

A. The parametric problem

We consider the parameter vector $\theta \equiv (q, b, \chi^A, \chi^B)$ at fixed total bare mass $M \equiv m_A + m_B$ (set to 1), with the spins carried as the Kerr-like *dimensionless* spins $\chi^X \equiv \mathbf{S}^X/m_X^2$, each a Cartesian 3-vector referred to the bare puncture mass m_X . The per-puncture momenta are not interpolation axes but are determined by the quasi-circularity condition of Sec. V A. The map to a slice is algebraic: $m_A = Mq/(1+q)$, $m_B = M/(1+q)$, and the Bowen–York spin that enters the analytic source is $\mathbf{S}^X = \chi^X m_X^2$. The interpolation axes are q , b , and the individual spin *components* χ_i^X , i.e. up to eight scalar axes for the full two-spin family.

Sampling the spins in χ_i^X keeps the parameter domain rectangular, because the physical Kerr bound $|\chi_i^X| \leq 1$ is independent of the component masses. The spin axes are therefore fixed independent of q and symmetric about zero. Each interpolation axis is finally mapped affinely onto $[-1, 1]$, the natural domain of the Chebyshev basis.

B. Value (Lagrange) collocation

Because the ABT coordinate system and collocation grid are identical at every parameter node, all solutions are represented on the same discrete function space. Each nodal value therefore corresponds to the same physical location in the compactified coordinate domain throughout parameter space. This fixed representation makes tensor-product interpolation over the parameters well-defined: the interpolant acts directly on corresponding nodal values, producing a surrogate field on the same spectral grid. The resulting surrogate inherits the smoothness of the underlying family of solutions and can be evaluated at arbitrary parameter values before being interpolated to arbitrary spatial coordinates using the coefficient-free spectral interpolation described in Sec. III A. The resulting parametric model is then,

$$\tilde{u}(\mathbf{x}; \theta) = \sum_{i_1=0}^{Q_1} \cdots \sum_{i_d=0}^{Q_d} U_{i_1 \dots i_d}(\mathbf{x}) \prod_{k=1}^d L_{i_k}(\theta_k), \quad (21)$$

where $U_{i_1 \dots i_d}$ is the high-fidelity solve stored at the parameter node $(\theta_{1,i_1}, \dots, \theta_{d,i_d})$ and L_{i_k} is the Lagrange cardinal function associated with the k -th parameter. Equation (21) defines the method: initial data at a new parameter value are obtained as a parameter-independent linear combination of the stored high-fidelity solutions.

We evaluate the cardinal functions in the numerically stable barycentric form [16],

$$L_{i_k}(\theta_k) \equiv \frac{\lambda_{i_k}/(\theta_k - \theta_{k,i_k})}{\sum_{j_k=0}^{Q_k} \lambda_{j_k}/(\theta_k - \theta_{k,j_k})}, \quad (22)$$

where $\lambda_{i_k} = (-1)^{i_k} \delta_{i_k}$, with $\delta_{i_k} = 1/2$ at the end-points and unity otherwise, so that $L_{i_k}(\theta_{k,j_k}) = \delta_{i_k j_k}$. The nodes θ_{k,i_k} are the affine images of the Chebyshev–Gauss–Lobatto points on $[\theta_k^{\min}, \theta_k^{\max}]$.

The tensor-product interpolant is evaluated by contracting one parameter axis at a time, requiring $O(\prod_k (Q_k + 1))$ operations per query. More important than the per-query cost is that the interpolant provides an accurate initial guess throughout parameter space. Consequently, the certified polish of Sec. IV C reaches tolerance in only two to four Newton iterations, compared with approximately seven iterations from a cold start. Because the parameter-to-solution map is analytic, the interpolation error decays exponentially in the number of parameter nodes, at a rate fixed by the nearest singularity in the complex parameter plane. This exponential convergence is demonstrated in Sec. V C.

C. Certified refinement

The surrogate of Eq. (21) supplies only a *guess*. We certify it with a few Newton steps of the genuine elliptic solver of Sec. III B. Taking the barycentric prediction as the initial iterate, $u^{(0)} = \mathcal{I}[u](\theta)$, we then apply the Newton iteration of Eq. (17), with the residual R and analytic Jacobian J of Eqs. (18,19). Because the guess is already close, a few steps drive the constraint residual to $\|R\|_\infty \leq 10^{-10}$ at *any* queried parameter, *independent* of the interpolation error. This is the certification guarantee; Sec. V D makes it quantitative for the shipped models.

Two quite different quantities measure how good a guess $u^{(0)}$ is: its *field* error $\|u^{(0)} - u_{\text{true}}\|_2 / \|u_{\text{true}}\|_2$, the ordinary distance between fields, which the interpolation accuracy controls; and the *constraint residual* $\|R(u^{(0)})\|_\infty$, which is what the Newton iteration (17) actually reduces. These are nearly *decoupled*. The residual operator differentiates the field (it contains the Laplacian, at scale $\propto b^{-2}$), so it amplifies precisely the small-amplitude, high-derivative content that the field norm barely registers: a guess can be excellent in field-norm yet violate the constraint by orders of magnitude more. Concretely, projecting the *exact* solution onto a smooth low-dimensional subspace leaves a field error $\sim 10^{-4}$ but a residual $\|R\|_\infty \sim \mathcal{O}(1)$, and it certifies in no fewer Newton steps than the crude barycentric guess. The step count is therefore set by $\|R(u^{(0)})\|_\infty$, not by field accuracy, with two consequences. First, refining the interpolant’s field accuracy is a weak lever on the online cost,

so the certified polish, which drives $\|R\|_\infty \rightarrow 0$ regardless of the guess, is the correct instrument. Second, one may discard the high-derivative content of the fields with no penalty, since the polish reconstructs it; this enables the reduced-basis re-encoding of Sec. IV H possible.

D. Differentiability

The parametric solver is differentiable with respect to the physical parameters: for any queried θ it exposes the sensitivities $\partial U/\partial\theta$ of the emitted field and, through any observable F constructed from it, the sensitivity $\partial F/\partial\theta$. That an interpolant is differentiable is unremarkable; the content of the claim is that these are derivatives of the *certified* initial data. The challenge is therefore not differentiating the interpolant, but preserving differentiability through the projection back onto the Einstein constraint manifold. Implemented in JAX [21] as a branchless barycentric map, the surrogate admits forward-mode automatic differentiation, which returns $\partial U/\partial\theta$ and $\partial F/\partial\theta$ directly, without numerical approximation.

At each node the tangent of the certified solution is available directly. Differentiating the vanishing residual identity $R(U(\theta), \theta) \equiv 0$ once gives the first-order forward-sensitivity equation

$$J \frac{\partial U}{\partial \theta_k} = - \frac{\partial R}{\partial \theta_k}, \quad (23)$$

with the same Jacobian $J = \partial R/\partial U$ (Eq. 19) that the certified solve has already factored, so each tangent costs one back-solve and no new factorization. The spin axes need a closed-form source derivative: χ enters the residual only through $\hat{A}^2$ via $S^X = \chi^X m_X^2$, so $\partial R/\partial \chi^X = \frac{1}{8}(\psi_{\text{BL}} + u)^{-7} m_X^2 \partial \hat{A}^2/\partial S^X$, which is node-diagonal and requires no finite differencing.

Two references verify the exposed sensitivities, and they play distinct roles (Appendix B). Second-order central finite differences of the same differentiable surrogate confirm the automatic differentiation itself. The load-bearing check is against the implicit-function-theorem tangent of Eq. (23), computed independently from the certified solve at the same parameter point: this is what establishes that the exposed gradient is the sensitivity of the certified datum rather than of the interpolant.

Equation (23) is itself checked against finite differences of the certified three-dimensional solve. The remaining agreement is limited by the interpolation accuracy of the derivative, which converges exponentially in the parameter order just as the field interpolation does, and is correspondingly weakest along the slowly converging b and q axes.

Certification and differentiability therefore coexist rather than compete: every returned datum satisfies the constraints to tolerance, and its exact dependence on the physical parameters is available at once. This is what separates these sensitivities from those of a bare interpolant or a neural-network surrogate, and what makes the gradient-based applications of Sec. VI possible.

E. Gradient-enhanced (Hermite) collocation

Every certified solution carries verified parameter tangents $U'_i \equiv dU/d\theta_k|_{\theta_i}$, produced during the offline build (Sec. IV F) for the cost of a single back-solve against the node's already-factored Jacobian (Sec. IV D). This derivative information has therefore already been paid for, and to interpolate only the stored field values U_i of Eq. (21) would discard it. Gradient-enhanced (Hermite) collocation reincorporates it into the surrogate, matching values *and* parameter-derivatives at each node.

The gain is worth harvesting only where interpolation converges slowly. The fast parameter directions already converge spectrally, so matching tangents helps most in the slow directions near the Bernstein walls quantified in Sec. V C. We therefore enhance only a subset $\mathcal{E}$ of the axes, and leave every other axis with its plain Lagrange cardinal L_{i_k} of Eq. (22). Along an enhanced axis $k \in \mathcal{E}$ that cardinal is replaced by the Hermite pair $(H_{i_k}, \hat{H}_{i_k})$, and when two axes are enhanced together consistency requires matching their mixed cross-derivative as well, giving the genuine tensor-product Hermite interpolant on the enhanced subspace. For the two enhanced axes used throughout this work, let i_A and i_B denote the two indices i_k with $k \in \mathcal{E}$, write $U \equiv U_{i_1 \dots i_d}(\mathbf{x})$ for the certified solve stored at the node, and suppress the cardinal arguments, $H_{i_A} \equiv H_{i_A}(\chi_y^A)$ and $\hat{H}_{i_B} \equiv \hat{H}_{i_B}(\chi_y^B)$. The surrogate of Eq. (21) then generalizes to

$$\tilde{u}(\mathbf{x}; \theta) = \sum_{i_1=0}^{Q_1} \dots \sum_{i_d=0}^{Q_d} \left(\prod_{k \notin \mathcal{E}} L_{i_k}(\theta_k) \right) \left[H_{i_A} H_{i_B} U + \hat{H}_{i_A} H_{i_B} \partial_{\chi_y^A} U + H_{i_A} \hat{H}_{i_B} \partial_{\chi_y^B} U + \hat{H}_{i_A} \hat{H}_{i_B} \partial_{\chi_y^A} \partial_{\chi_y^B} U \right], \quad (24)$$

whose four terms are the value, the two certified first tangents, and the single bilinear cross tangent stored at the node (Sec. IV D). Each non-enhanced axis carries only its Lagrange cardinal, so restoring L on the two enhanced

axes and dropping the tangents recovers the value interpolant of Eq. (21). The osculatory Hermite cardinals are built from the same barycentric Lagrange cardinal L_i of

Eq. (22),

$$H_{i_k}(\theta) \equiv [1 - 2L'_{i_k}(\theta_{i_k})(\theta - \theta_{i_k})] L_{i_k}(\theta)^2, \quad (25)$$

$$\hat{H}_{i_k}(\theta) \equiv (\theta - \theta_{i_k}) L_{i_k}(\theta)^2, \quad (26)$$

with $L'_i(\theta_i) = \sum_{j \neq i} (\theta_i - \theta_j)^{-1}$ the diagonal of the barycentric differentiation matrix. They satisfy $H_{i_k}(\theta_{j_k}) = \delta_{i_k j_k}$, $H'_{i_k}(\theta_{j_k}) = 0$, $\hat{H}_{i_k}(\theta_{j_k}) = 0$, and $\hat{H}'_{i_k}(\theta_{j_k}) = \delta_{i_k j_k}$, so the interpolant reproduces the certified value *and* tangents at every node by construction, and the exposed gradient at the nodes equals the certified-solve sensitivity exactly rather than the derivative of a value interpolant. Because the parameter-to-solution map is analytic, matching the tangent raises the per-node convergence order, so the same warm-start quality is reached with fewer expensive elliptic solves.

For both shipped models the enhanced set is the

pair of aligned spin components, $\mathcal{E} = \{\chi_y^A, \chi_y^B\}$, identified as the slowest axes in Sec. VB. The remaining axes carry plain Lagrange cardinals: b and q in the four-dimensional aligned-spin model $\theta = (b, q, \chi_y^A, \chi_y^B)$, and those two together with the four in-plane components $\chi_x^A, \chi_z^A, \chi_x^B, \chi_z^B$ in the eight-dimensional general-spin model $\theta = (b, q, \chi^A, \chi^B)$. The bracket of Eq. (24) is therefore identical in the two models; only the number of Lagrange factors differs.

Restricting enhancement to $\mathcal{E}$ also keeps the stored-derivative count small: a full tensor-product Hermite over *all* d axes would carry mixed partials up to order d (2^d fields per node), whereas Eq. (24) carries only the $2^{|\mathcal{E}|} = 4$ derivatives internal to the enhanced subspace. The cross term is as cheap as the first tangents: differentiating the vanishing residual identity $R(U(\theta), \theta) \equiv 0$ twice gives the second-order forward-sensitivity equation

$$J \frac{\partial^2 U}{\partial \theta_k \partial \theta_{k'}} = - \left[\frac{\partial^2 R}{\partial \theta_k \partial \theta_{k'}} + \frac{\partial^2 R}{\partial \theta_k \partial U} \frac{\partial U}{\partial \theta_{k'}} + \frac{\partial^2 R}{\partial \theta_{k'} \partial U} \frac{\partial U}{\partial \theta_k} + \frac{\partial^2 R}{\partial U^2} \left[\frac{\partial U}{\partial \theta_k}, \frac{\partial U}{\partial \theta_{k'}} \right] \right], \quad (27)$$

with the *same* factored Jacobian $J = \partial R / \partial U$ as the first tangent (Eq. 19), i.e. one further back-solve and no new factorization. Because the spins enter the residual only through $\hat{A}^2$ (Sec. IVD), every term on the right is a closed-form, node-diagonal function of the source and its first two spin-derivatives, so no finite differencing is needed. Supplying this cross term closes the off-node cross-curvature between the two enhanced axes that a gradient-only construction, carrying the two first tangents but not their cross, leaves approximated; it lowers the joint held-out error (Sec. VD). Enlarging $\mathcal{E}$ beyond two axes would call for the higher mixed partials of the correspondingly larger tensor product, and we do not pursue it: Sec. VD finds that a larger enhanced set degrades the model rather than improving it.

The gradient-enhanced surrogate is thus more accurate at no additional expensive cost: it reuses derivative information already produced by the offline build, introduces no new elliptic solves, and concentrates its accuracy gain along the slowly converging spin directions where interpolation is hardest. **It only requires more memory to store the additional gradient information (see figure memory).** Certification is unaffected: the enhanced surrogate remains only a warm start, refined to tolerance by the certified Newton step of Sec. IVC.

F. Building the grid: warm-started continuation

The construction of the parametric model is independent of the underlying elliptic solver. It requires only a routine `solve( $\theta$ , guess)` that computes the solution at a parameter point θ from an initial guess and, option-

ally, the parameter tangent $dU/d\theta$. The interpolation nodes are populated by traversing the tensor grid in a boustrophedon (reflected-serpentine) order, so that successive nodes only differ by a single parameter increment. Each Newton solve can therefore warm-start from the converged solution at a neighbouring parameter point, while only the first solve requires a cold start.

G. Sparse grids for higher dimensions

The full tensor-product interpolant of Eq. (21) requires one expensive elliptic solve per parameter node, so its offline cost $\prod_k (Q_k + 1)$ grows exponentially with the number of parameters d . This scaling is unnecessary because the parameter-to-solution map is analytic. Analytic dependence makes the interpolation error decay geometrically with the per-axis resolution (Sec. IV B), so refining every direction to high order is wasteful: the accuracy is dominated by a few finely resolved parameters rather than by resolving all of them simultaneously. A *Smolyak sparse grid* [13, 24] retains only these dominant tensor-product contributions.

Index each axis by a resolution *level* l_k , and let I_{l_k} be the one-dimensional interpolation operator on the $m(l_k)$ nodes of that level, with $m(0) = 1$ and $m(l) = 2^l + 1$ Clenshaw–Curtis doubling nodes¹. These node sets are

¹ The Clenshaw–Curtis nodes $\cos(\pi j / 2^l)$, $j = 0, \dots, 2^l$, are the extrema of the Chebyshev polynomials; taking $2^l + 1$ of them (and thus doubling the number of subintervals each level) makes

nested, and the levels telescope through the hierarchical surpluses $\Delta_l \equiv I_l - I_{l-1}$ (with $\Delta_0 \equiv I_0$), so that $I_L = \sum_{l \leq L} \Delta_l$. For an analytic map the surpluses decay geometrically in l , through the same Bernstein-ellipse mechanism that governs the interpolation error (Sec. VC).

In d dimensions the interpolant telescopes into a sum of mixed surpluses over all levels, $\bigotimes_k I_L = \sum_l \bigotimes_k \Delta_{l_k}$, recovering the $O(Q^d)$ cost. For an analytic map these mixed contributions decay geometrically in the total level $|\mathbf{l}|_1 = \sum_k l_k$, so resolving fine structure in several parameters at once is exponentially redundant. Retaining the *simplex* $|\mathbf{l}|_1 \leq \ell$ captures nearly all of the interpolation accuracy and discards only exponentially small contributions. The retained surpluses recombine into the Smolyak combination technique,

$$\mathcal{I}_{\text{sparse}}[u] = \sum_{|\mathbf{l}|_1 \leq \ell} c_l \bigotimes_{k=1}^d I_{l_k}[u], \quad (28)$$

with the integer coefficients

$$c_l = (-1)^{\ell - |\mathbf{l}|_1} \binom{d-1}{\ell - |\mathbf{l}|_1}, \quad (29)$$

nonzero only for $\ell - d + 1 \leq |\mathbf{l}|_1 \leq \ell$. Because the one-dimensional nodes are nested, the union of these subgrids is the sparse grid, and every node is solved only once.

The sparse grid replaces the exponential offline solve count with one that grows as $O(m(\log m)^{d-1})$ nodes versus $O(m^d)$, i.e. polynomially in the per-axis resolution and near-linearly in the dimension, while retaining nearly the full tensor-product accuracy. This scalability makes the higher-dimensional general-spin family tractable, where the full tensor grid is not, and the saving in offline solve count is quantified in Sec. VD.

Sparse grids alter only the offline sampling strategy: each subgrid interpolant in Eq. (28) is itself an instance of the surrogate of Eq. (21), so the online interpolation, the certified Newton refinement of Sec. IV C, and the differentiability of Sec. IV D carry over unchanged. When the parameters differ in difficulty, as the slowly converging spin axes of Sec. VB do, the isotropic simplex can be replaced by a *dimension-adaptive* downward-closed index set [13] grown greedily along the axes whose surpluses remain large, a refinement we leave to future work. In every case the sparse grid reduces the number of expensive offline solves without weakening the certification guarantee or the exact differentiability that define the method.

level l a subset of level $l+1$, i.e. nested, since $\cos(\pi j/2^l) = \cos(\pi(2j)/2^{l+1})$, such that the even nodes of level $l+1$ are equal to the nodes of level l .

H. Reduced-basis re-encoding

Because the model *is* its stored nodal solves (Eq. 21), its memory grows linearly in the node count, which becomes a liability in high dimension. However, since the parameter-to-field map is smoothly low rank, the corpus admits a lossless reduced-basis re-encoding.

We compress by *proper orthogonal decomposition* (POD, equivalent to principal-component analysis) [25, 26]. Stacking the N stored fields as rows of a matrix and subtracting their mean $\bar{u}$, a singular-value decomposition yields orthonormal *spatial* modes $\Phi = [\phi_1, \phi_2, \dots]$ ordered by singular value σ_k ; each stored field can then be written as $u = \bar{u} + \Phi c$ with coefficients $c = \Phi^\top (u - \bar{u})$. The singular values decay geometrically, because the map is analytic in θ , the same analyticity that governs the interpolation error (Sec. VC). Keeping the leading r modes, we store Φ together with the length- r coefficient vector at each node, i.e. r numbers per node in place of the full field, and, at query time, interpolate the *coefficients* with the identical barycentric/Smolyak machinery of Eq. (21) and decode $u(\theta) = \bar{u} + \Phi c(\theta)$.

Two properties make this a re-encoding rather than a lossy approximation. First, interpolation is linear, so interpolating the projected coefficients is *identical* to projecting the full-field interpolant onto the r leading modes: the compressed model is exactly the rank- r truncation of Eq. (21), differing from it only by a controllable truncation tail. Second, and this is why even that tail is harmless, the certified polish of Sec. IV C re-solves the constraint from the decoded guess, reconstructing precisely the high-derivative content the truncation discards; by the residual/field-accuracy decoupling above, the truncation changes the warm start's field error but not its Newton-step count. The differentiable evaluation of Sec. IV D also survives: the parameter Jacobian of the compressed model is the full Jacobian projected onto Φ , which loses a negligible fraction of the gradient. The decomposition is computed once from the existing corpus, with no new solves; Sec. VF reports the compression achieved on the quasi-circular demonstration models.

V. QUASI-CIRCULAR PARAMETRIC MODELS

We now assemble the two headline models, both *quasi-circular* orbiting initial data built by the identical machinery over the production box $b \in [2, 7] M$ (puncture separations $D = 2b$ up to $14M$), $q \in [1, 3]$, and dimensionless spins with every component in $[-0.99, 0.99]$, at fixed total mass $M = 1$: a four-dimensional *aligned-spin* model $\theta = (b, q, \chi_y^A, \chi_y^B)$ and the full eight-dimensional *general-spin* model $\theta = (b, q, \chi^A, \chi^B)$ (Sec. VE), whose in-plane spin axes are centred on zero so the aligned model nests, node-for-node, inside it. **[TODO: production box — both stated ranges disagree with the canonical definition in pipeline/production_box.py. (i) The sep-**

aration is now $b \in [3, 10] M$, i.e. $D = 2b \in [6, 20] M$, not $b \in [2, 7] M$ “up to $14 M$ ”; the same stale range appears in the caption of Fig. 4. (ii) The spin half-width is $\text{CHI_MAX} = 0.9$, not 0.99 — set deliberately to sit inside the Bowen–York horizon-spin ceiling $\chi_{\text{hor}} \approx 0.93$, above which the horizon spin saturates. Which value the shipped corpora were built at is an author decision, not an editorial fix; see also the physical ranges quoted in Sec. VII.] Each is built at Smolyak level $\ell = 5$ (Sec. IV G), gradient-enhanced on the spin axes (Sec. IV E), and reduced-basis compressed (Sec. IV H); this gradient-enhanced, compressed model is the shipped artifact, with the value-only surrogate its $\mathcal{E} = \emptyset$ baseline.

The accuracy metric throughout is the *held-out* interpolation error: at a parameter point that is *not* a collocation node we compare the interpolant to a direct elliptic solve on the *same* frozen spatial grid ($N_A = 44$, $N_B = 32$, $N_\phi = 8$) and take the largest nodal difference between the two fields. Because both are represented on the same grid, the spatial discretization error cancels and what remains is the pure parameter-interpolation error. We report it in two ways. The *per-axis* error sweeps a single parameter through 21 held-out points, at each of 100 random base points for the remaining axes, and so isolates the convergence rate of that one axis. The *joint* held-out error instead draws 1000 parameter points uniformly at random from the whole box, so that every axis is off-node at once; it is the error encountered at an arbitrary query, and it is governed by the slowest-converging axis. Both are distributions over the sampled points, which we report as a median with its extremes.

A. The quasi-circular momentum condition

What makes the family *orbiting* rather than a free momentum scan is that the per-puncture momenta are not free parameters but a deterministic function of (b, q, spins) , fixed by a post-Newtonian quasi-circularity condition: a closed-form tangential momentum together with a small radial component that vanishes in the circular limit, so every node is a physical orbiting configuration and no free momentum axis is carried. The tangential momentum lies along x , off the separation (z) axis, so the orbital angular momentum points along $+y$ and the data are non-axisymmetric. We fix the momenta by the standard puncture quasi-circularity procedure [27, 28]. Defining $M \equiv m_A + m_B$, $\mu \equiv m_A m_B / M$, $\nu \equiv \mu / M$, coordinate separation $D \equiv 2b$, and $x \equiv M/D$, the per-puncture *tangential* momentum is the non-spinning post-Newtonian series through 3PN [28],

$$\begin{aligned} \frac{P_t}{\mu} = & x^{1/2} + 2x^{3/2} + \frac{1}{16}(42 - 43\nu)x^{5/2} \\ & + \frac{1}{128}(480 + (163\pi^2 - 4556)\nu + 104\nu^2)x^{7/2}. \end{aligned} \quad (30)$$

Note that the leading term is the Newtonian value $P_t \rightarrow \mu\sqrt{M/D} = \mu\sqrt{M/2b}$. We add the leading (1.5PN) spin-orbit correction [29], which depends only on the spin projections $\chi_{\parallel}^X$ onto $\hat{\mathbf{L}}$; in-plane spin components first enter at higher PN order and are neglected. Converted to the convention used throughout this paper² ($q = m_A/m_B \geq 1$, A the larger hole; $\chi_{\parallel}^X \equiv (\mathbf{S}^X \cdot \hat{\mathbf{L}})/m_X^2$ the dimensionless spin projected on the orbital angular momentum, i.e. the aligned component), it reads

$$\frac{P_t^{\text{SO}}}{\mu} = \frac{2}{3(1+q)^2} [q(4q+3)\chi_{\parallel}^A + (3q+4)\chi_{\parallel}^B] x^2. \quad (31)$$

The *radial* component is the leading radiation-reaction (Peters) momentum $P_r = \mu |\dot{D}|$ with the quadrupole inspiral rate $\dot{D} = -\frac{64}{5}\mu M^2/D^3$ [28, 30], i.e.

$$\frac{P_r}{\mu} = \frac{64}{5} \frac{\mu M^2}{D^3}, \quad (32)$$

so $P_r/P_t \propto \nu(M/D)^{5/2} \rightarrow 0$ as $b \rightarrow \infty$, yielding the circular limit. The per-puncture momenta are then $\mathbf{P}_A = (+P_t, 0, -P_r)$, $\mathbf{P}_B = (-P_t, 0, +P_r)$, with zero net linear momentum and orbital angular momentum $2bP_t$ along $+y$; only the aligned (S_y) spin component enters Eq. (31), so in-plane spins leave the momenta unchanged at this order. Both limits and the resulting angular momentum are validated against the closed form and TwoPunctures in Appendix A.

B. Convergence across the parameter space

Sweeping one parameter at a time, every axis converges geometrically (Fig. 1). The separation b and mass ratio q are the fastest; the spins are somewhat slower, but all eight rates lie within a narrow band, and the two holes’ spin axes converge at comparable rates. Gradient enhancement (Sec. IV E) roughly doubles every rate for the price of one extra back-solve per node against the node’s already-factored Jacobian. Because the joint held-out error is governed by the slowest of these axes, it is the spin directions that set the offline cost of the model (Sec. V D).

C. Analyticity limits of the parameter map

The convergence rate of a spectral parameter interpolant is set by the distance from the sampling interval to the nearest singularity of the parameter-to-solution

² Healy *et al.* [29] write this correction with $q = m_2/m_1 \leq 1$ (body 1 the larger hole); Eq. (31) is the algebraically identical form obtained by substituting $q \rightarrow 1/q$ and relabelling $(1, 2) \rightarrow (A, B)$. It is invariant under $q \rightarrow 1/q$ with $\chi_{\parallel}^A \leftrightarrow \chi_{\parallel}^B$, as it must be.

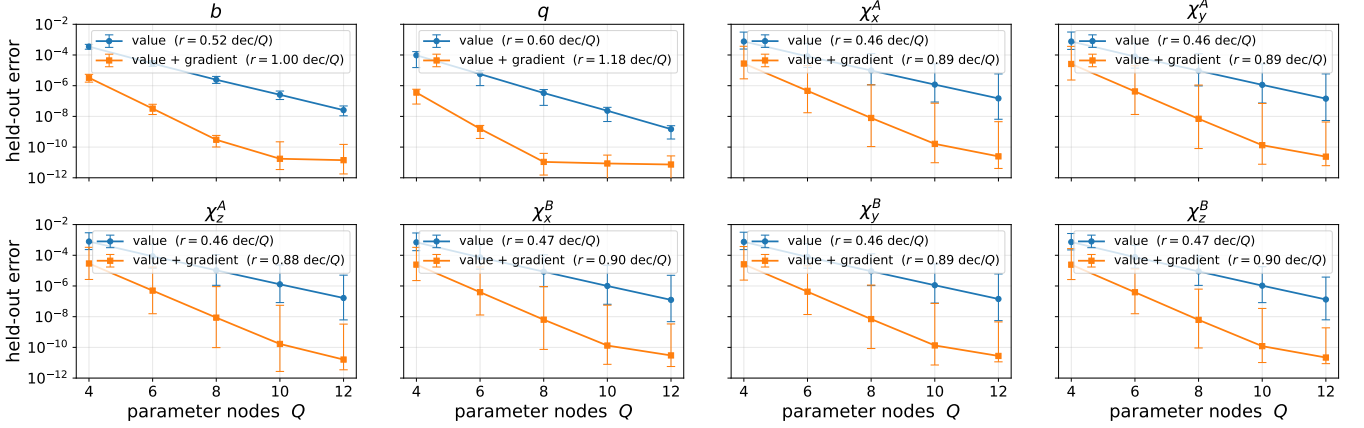


FIG. 1. **Per-axis convergence of the quasi-circular models.** Held-out interpolation error against the number of parameter nodes for each of the eight axes (b, q, χ^A, χ^B); the first four are the four-dimensional aligned-spin model. The value-only surrogate (“value”) is compared with the gradient-enhanced one (“value+gradient”), which also matches the certified parameter tangent at each node. A one-parameter sweep carries no mixed tangent, so the cross term of Eq. (24) does not enter here. Markers are medians over 100 random base points for the non-swept axes, with whiskers spanning the best–worst range. Every axis decays geometrically, at the rates fit from the median curves, and matching the tangent roughly doubles each rate.

map in the complex plane, through the *Bernstein-ellipse* convergence bound (Theorem 8.2 of Ref. [16], going back to [31]). We map the sampling interval $[\theta_{\min}, \theta_{\max}]$ of a parameter θ affinely onto $[-1, 1]$ by $\theta \mapsto \xi = (2\theta - \theta_{\max} - \theta_{\min})/(\theta_{\max} - \theta_{\min})$, and consider the *Bernstein ellipses* E_ϱ , which are defined as the ellipses in the complex ξ -plane with foci at ± 1 whose semi-major and semi-minor axes sum to $\varrho > 1$. If the parameter-to-solution map is analytic inside E_ϱ , its Q -node Chebyshev interpolant converges geometrically, with error $\sim \varrho^{-Q}$, i.e. a rate of $\log_{10} \varrho$ decades per node. The operative ϱ is fixed by the *largest* ellipse free of singularities, so the nearest singularity of the map lies *on* E_ϱ . A singularity at the mapped point ξ_* sits on the ellipse of parameter $\varrho = |\xi_* + \sqrt{(\xi_*)^2 - 1}|$, which for a real singularity outside the interval reduces to $\varrho = |\xi_*| + \sqrt{(\xi_*)^2 - 1}$. Hence, a singularity in parameter space close to the sampled box $[\theta_{\min}, \theta_{\max}]$ will yield a ϱ just above unity and thus slow convergence, while a distant singularity will yield a large ϱ and fast convergence.

We read this estimate *backwards* to locate the wall. From the slope of the held-out error against the node count Q we measure the geometric convergence rate $r \equiv \log_{10} \varrho$, invert it to $|\xi_*| = \frac{1}{2}(\varrho + \varrho^{-1})$, and map ξ_* back through the affine map, taking the real root outside the sampled box on the physically singular side (small b toward merger for the separation; large $|\chi|$ toward the extremal limit for the spins), to read off the inferred singularity location θ_* . Sweeping the sampling range then diagnoses the *character* of the wall. A genuine real branch point stays pinned at a fixed θ_* as the range changes. A complex-conjugate pair off the real axis has no real singularity at all, yet the one-parameter fit still returns a real θ_* , the real point lying on the same Bernstein ellipse as the true complex pair, which *marches* outward as the

range grows. The legends of Fig. 2 report this inferred θ_* for each sampling range alongside the fitted rate; whether θ_* holds still or recedes as the range grows separates the two wall types below.

The quasi-circular family has two walls of opposite character (Fig. 2). Both sweeps are run at equal mass, with the remaining parameters at their symmetric values. **[TODO: walls rebuild — “two walls” becomes THREE (a mass-ratio wall now sits between them in Fig. 2); and “both sweeps are run at equal mass” cannot cover the q sweep, whose swept axis is the mass ratio (it holds $b = 3.5$, spins zero). Rewrite this paragraph once block Q_wall_q lands.]**

The *merger* wall, in the separation b , is hard. As the sampling interval is pushed toward coincidence the rate degrades smoothly and tracks the Bernstein prediction for a singularity fixed at $b = 0$, the inferred b_* staying at the merger throughout. This resolves the concern (specific to the orbiting family) that the $b \rightarrow 0$ limit is *doubly* singular: the punctures coincide *and* the post-Newtonian tangential momentum $P_t \propto b^{-1/2}$ diverges, yet the extra branch does not *relocate* the wall off $b = 0$. Its only signature is a mild undershoot of the measured rate below the pure-coincidence prediction at the smallest $b_{\min}$, i.e. the momentum divergence makes the wall marginally *stronger*, not closer.

The *spin* wall, in χ , is by contrast soft. The measured rate degrades only mildly as the range grows. Here, the range was deliberately extended past the production box to locate the wall and the inferred nearest *real* singularity marches outward, staying beyond the sampled range throughout, which is the signature of a complex-conjugate pair off the real axis rather than of a real wall. That nearest feature is already super-extremal, well past

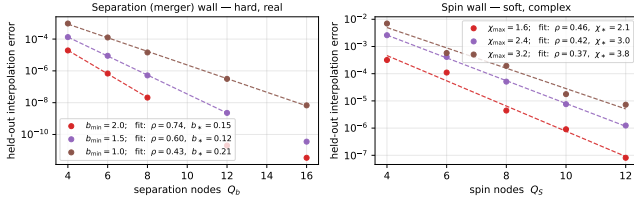


FIG. 2. **The analyticity walls of the quasi-circular family.** Held-out interpolation error against the number of nodes on the swept axis, for several sampling ranges, with the geometric fit $\varepsilon \approx A 10^{-\rho Q}$ dashed over the fitted window, which ends where the error saturates on the float64 floor. Legends give, per range, the fitted rate ρ (decades/node) and the inferred nearest real singularity θ_* . *Top (separation b):* the rate degrades as the interval is pushed toward coincidence while b_* stays pinned at the merger — a hard, real wall. *Middle (mass ratio q):* [pending the q -wall measurement]. *Bottom (spin χ):* the rate degrades only mildly and χ_* marches outward, beyond the sampled range — a complex-conjugate pair off the real axis, hence soft (Sec. V C). **[TODO: on rebuild — write the middle panel’s sentence and, if the measurement warrants it, the wall’s character into its panel title. Every number here also changes: the current curves are the superseded narrow box $b \in [1.5, 4]$ in the old bare-spin (S_{Ay}, S_{By}) parameterization.]**

the Cook–York horizon-spin ceiling [32], so spin is a benign axis with no sharp singularity near the physical box. What sets the convergence rate is therefore the *distance* of the nearest singularity and not its hardness: the family’s only hard wall lies on its fastest axis, while the slowest axes carry only soft, receding ones. **[TODO: load-bearing claim — re-derive, do NOT patch. The fastest/slowest framing of this sentence presumes exactly two walls. The mass-ratio axis is neither the fastest nor the slowest, so a pinned q_* would place a second hard wall on a middling axis and dissolve the dichotomy; a marching q_* preserves it but adds a third case. The paragraph for the q wall itself is also still to be written, and its conclusion is the measurement’s to make: q_* pinned near 4.1 across the three ranges means $q_{\max} = 3$ is near a hard limit and the boxes cannot widen in q ; q_* marching means a complex pair, and $q_{\max}$ could rise to cover the asymmetric mass ratios. Under a real-singularity model the predicted rates are 0.600/0.375/0.170 at $q_{\max} = 3.0/3.5/4.0$.]**

D. Cost, memory, and certified residuals

The offline cost of a full tensor grid scales as $\prod_k (Q_k + 1) = O(Q^d)$; a sparse (Smolyak) grid [13, 24] breaks this and is what makes both the four-dimensional and the eight-dimensional quasi-circular models (Sec. V E) tractable. At isotropic Smolyak level $\ell = 5$ the four-dimensional model is 1105 solver nodes and

the eight-dimensional one 15,713, and the nested lower levels $\ell = 1 \subset \dots \subset 5$ are fully contained in each corpus, so no lower level costs additional solves. The offline saving is decisive: from the measured per-axis rates, a tensor grid built to a *bare* held-out accuracy of 10^{-6} – 10^{-9} would cost two to four orders of magnitude more solves — an accuracy the certified construction never needs, because the sparse grid already supplies a Newton-basin warm start everywhere in the box and the polish carries the correctness (the residual/field decoupling of Sec. IV C). Even at matched per-axis resolution the comparison is lopsided: a full tensor grid at the four-dimensional sparse grid’s finest level (33 nodes per axis) would be $\sim 10^3 \times$ the sparse node count.

Evaluating the interpolant is cheap. A bare query is a single contraction over the stored nodal solves, with no Newton iteration and no factorization, and costs only a small fraction of one Newton–Krylov step of the elliptic solve, and less still after the reduced-basis re-encoding of Sec. V F. The warm start it provides is therefore essentially free relative to the solve it seeds. A certified query is that same warm-started solve, and its saving over a cold solve is the reduced step count: two to four polish steps rather than the ~ 7 a cold start needs. The practical pattern is thus to bulk-generate initial data with bare queries and certify only the configurations one actually evolves. Because the model is nothing but its stored nodal solves, its memory is *linear* in the node count.

Enlarging the enhanced set is counterproductive. Because the enhanced subspace is where the extra back-solves are spent, we tested whether $\mathcal{E}$ should be enlarged beyond the two aligned-spin axes. Extending it to all four axes of the aligned-spin box at the same level degrades the held-out median by two orders of magnitude relative to doing nothing at all, and completing the pairwise-bilinear crosses does not repair it. The degradation is not an artifact of the reduced-basis compression, the cross layer, or the one-dimensional Hermite operators, each of which we checked in isolation: swept one axis at a time, Hermite beats Lagrange on *every* axis. What correlates is instead the *number* of simultaneously enhanced axes — enhancing any single axis improves on value-only, two degrade, and four degrade badly — consistent with cancellation in the Smolyak combination being governed by the one-dimensional operator norms, which multiply over the enhanced dimensions while the accuracy gain enters only once per axis. We have not established that mechanism directly. The practical consequence is that $\mathcal{E}$ should be kept small and spent on the slowest axes, and that the shipped $\mathcal{E} = \{\chi_y^A, \chi_y^B\}$, completed to full bilinear, is at or near the optimum of the configurations we measured.

E. The eight-dimensional general-spin model

Promoting the two aligned spins to full vectors takes the model to eight dimensions, $\theta = (b, q, \chi^A, \chi^B)$, where the tensor grid’s Q^d cost makes the Smolyak construction

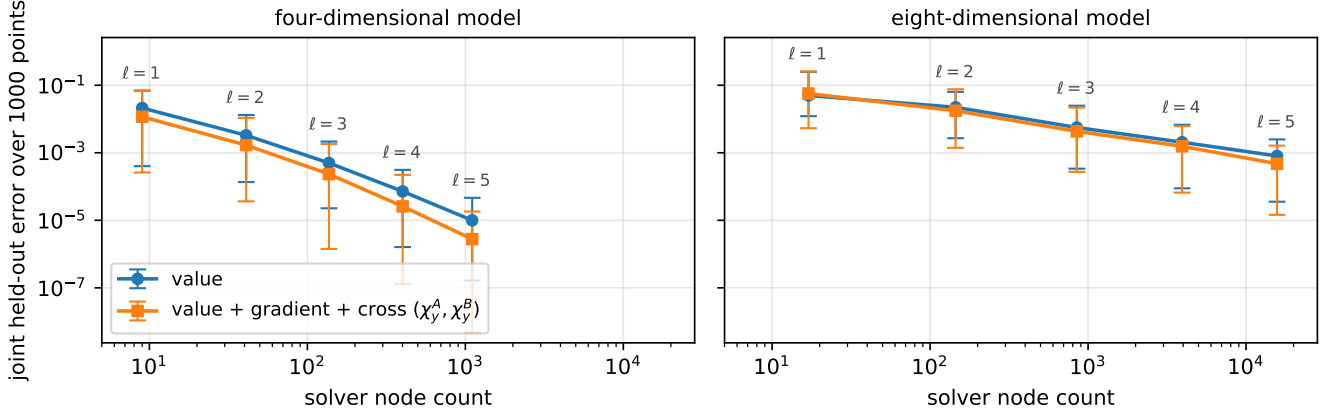


FIG. 3. **Joint held-out convergence.** Joint held-out error over 1000 random off-node points against solver node count (logarithmic, shared between panels), with the Smolyak level $\ell = 1$ –5 labelled at each point, shown as the best, median, and worst of the distribution; every level is served from the committed corpus at the node counts shown, with no new solves. *Left (four-dimensional model, $(b, q, \chi_y^A, \chi_y^B)$):* the value-only sparse interpolant (“value”) against the value+gradient+cross (full-bilinear Hermite–Smolyak) model enhanced on the two spin axes, which is more accurate at every level and reaches a $\sim 35\times$ deeper best-case floor at the same node count, from the node tangents alone. *Right (eight-dimensional general-spin model):* in preparation. The certified polish then drives every point to $\|R\|_\infty \leq 10^{-10}$ (Fig. 4).

of Sec. IV G not merely advantageous but necessary. The method is unchanged: with the in-plane spin axes centred on zero, the four-dimensional aligned model nests, node-for-node, inside the eight-dimensional one, and the $\ell = 5$ grid is 15,713 solver nodes, reusing the entire aligned corpus and adding only the genuinely-new preprocessing nodes. Its in-plane spin axes converge like the aligned ones, the two holes at comparable rates (Fig. 1), and the in-plane spin wall is again soft, its nearest real singularity receding as the fit range grows, so precession introduces no new hard wall. At this dimension it is the polish, not the bare sparse interpolant, that carries the accuracy: the 1000-point certified-polish statistics for both $\ell = 5$ models are reported in Fig. 4. The gradient-enhanced corpus reduced-basis compresses by a factor of 30 with no loss of certification or differentiability (Sec. V F).

F. Reduced-basis compression of the model

The corpora are correspondingly large: the gradient-enhanced four-dimensional $\ell = 5$ corpus (values, the d node tangents, and the bilinear cross tangent) is 583 MiB and the eight-dimensional one 13.5 GiB. This is a distribution liability for an artifact meant to be hosted and cited, and it is entirely removable by the reduced-basis (POD) re-encoding of Sec. IV H: an SVD of the stored fields yields orthonormal spatial modes Φ , of which we keep the leading r and interpolate the length- r coefficient vector at each node in place of the full field. The singular values decay geometrically (the map is analytic in θ , as in Sec. V C): for the four-dimensional model the rank- r reconstruction error of the corpus falls below 10^{-6} by $r = 87$ modes, and because the value and its param-

eter tangents *share* one spatial basis, the enhanced corpus needs only some 12% more modes than the value-only one at the same tail. The rank at a fixed accuracy is set by the intrinsic dimension of the solution manifold, not by the sampling density: it is the counterpart to Sec. IV G along the orthogonal axis — the Smolyak level fixes the parameter resolution and the offline solve count, POD the spatial rank, and the two are independent.

Figure 5 shows the resulting memory–accuracy trade-off. At the shipped ranks the four-dimensional model shrinks by a factor of 51 (583 MiB to 11.5 MiB) and the eight-dimensional one by 30 (13.5 GiB to 461 MiB), and per-query evaluation, now a barycentric sum over r coefficients rather than the full field, drops to a few milliseconds. The certified constraint residual, the number of Newton steps needed to reach it (Sec. IV C), and the verified $\partial \text{ID} / \partial \theta$ (Sec. IV D) are all preserved: the mode projection loses only the truncation-tail fraction of the gradient, and the compressed model reproduces the truncated interpolant to that tail and reloads as a standalone predictor. The decomposition is computed once from the existing corpus, in minutes and with no new solves.

Three further compressions stack on top, all computed from the same corpus with no new solves, and all preserving the certified Newton-step count and the parameter gradient: single-precision storage of the spatial modes, an anisotropic Smolyak grid weighted toward the slowly-converging spin axes, and a per-Fourier-mode basis that stores the dominant $m = 0$ block in its compact form rather than the $8\times$ -redundant nodal one. Together they reduce even the eight-dimensional model several-fold further.

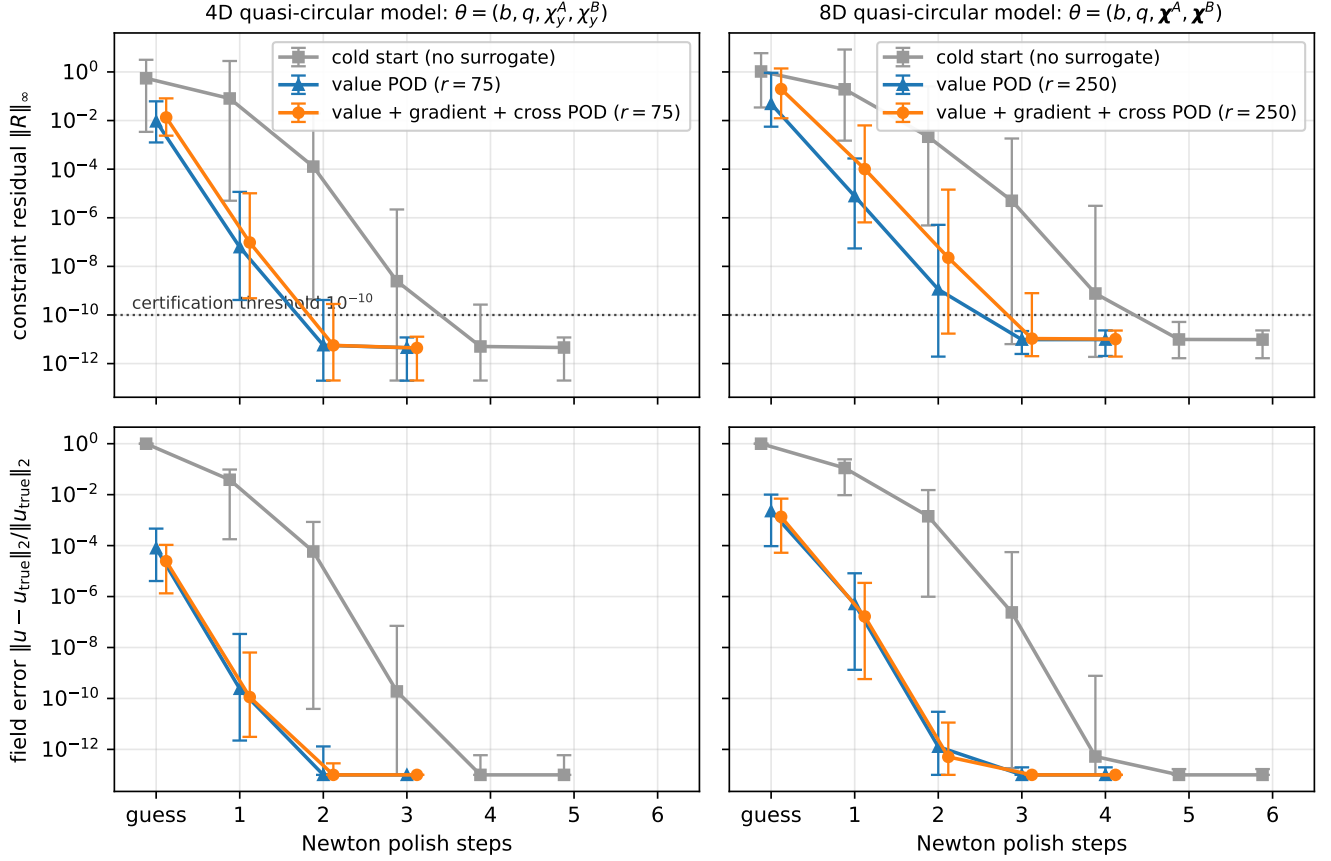


FIG. 4. **Certified refinement.** Constraint residual $\|R\|_\infty$ (top row) and relative field error $\|u - u_{\text{true}}\|_2 / \|u_{\text{true}}\|_2$ (bottom row) over 1000 uniformly-random off-node points of the quasi-circular models (χ , $b \in [2, 7]$, Smolyak $\ell = 5$; left column four-dimensional aligned-spin, right column eight-dimensional general-spin), before the polish (“guess”) and after each Newton step. Each curve is the median with min–max whiskers over the 1000 points, the field error measured against the converged certified iterate. Grey: a surrogate-free *cold start* from the solver’s zero-field iterate, whose bare residual is $\mathcal{O}(1)$. Blue: the value-only surrogate (“value”); orange: the shipped gradient-enhanced surrogate (“value+gradient+cross”), both POD-compressed to the same rank so that the two differ only in the gradient enhancement. Both surrogate warm starts certify *every* point to $\|R\|_\infty \leq 10^{-10}$ in at most four steps, about two ahead of the cold start, *independent* of the interpolation error. They are already field-accurate at the guess even though the residual there is orders of magnitude larger — the decoupling of Sec. IV C. [TODO: stale box — “ $b \in [2, 7]$ ” above is the superseded separation range; the production box is $b \in [3, 10] M$ (see the marker in Sec. V).]

G. External validation

The underlying initial data reproduce the established TwoPunctures [23] solver to spectral accuracy on the most demanding, genuinely non-axisymmetric configurations, and they are evolution-ready: on an evolution grid their finite-difference constraint violation is indistinguishable from TwoPunctures-sourced data. Because certification is an *internal* guarantee (Sec. IV C), this agreement is credibility rather than the correctness claim itself; Appendix A gives the full comparison.

VI. APPLICATIONS

The two payoffs of the method are that it is differentiable and constraint-certified, and we demonstrate both in two applications: certified parameter targeting on the flagship four-dimensional quasi-circular model, and eccentricity reduction on a purpose-built two-dimensional (b, P_t) model. Each targets a standard numerical-relativity initial-data workflow, and in each the honest cost metric is the number of *certified* elliptic solves.

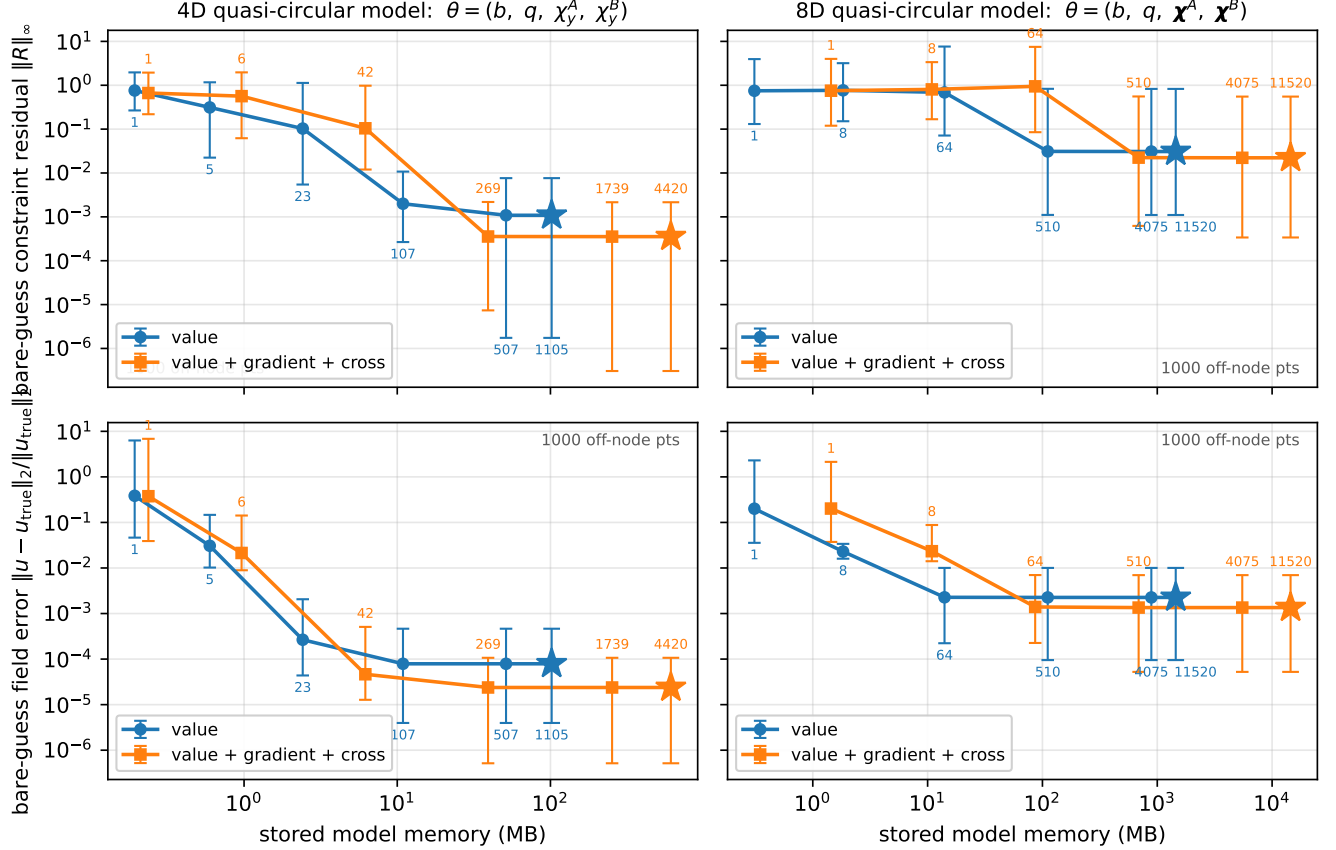


FIG. 5. **Reduced-basis (POD) compression versus stored memory.** Constraint residual $\|R\|_\infty$ of the bare barycentric guess (top row) and the relative field error $\|u - u_{\text{true}}\|_2 / \|u_{\text{true}}\|_2$ of the *same* guess against the certified solve (bottom row), as the POD truncation rank r is swept. Median line with min–max whiskers over 1000 uniformly-random off-node query points; each point is labelled by r and $\star$ marks the full-rank model. Every panel carries the same two families: *blue*, the value-only surrogate (“value”); *orange*, the shipped gradient-enhanced one (“value+gradient+cross”), which stores the node tangents and the bilinear cross tangent of Eq. (24) alongside the value and therefore sits at larger memory for the same rank. Columns share the memory axis: *left*, the four-dimensional quasi-circular model; *right*, the eight-dimensional general-spin model (Sec. V E), whose field-error sweep is pending. Compressing raises the bare-guess residual, but this warm-start quality is irrelevant to correctness: the certified polish (Sec. IV C) drives every query to $\|R\|_\infty \leq 10^{-10}$ in the same Newton-step count regardless. The field error meanwhile *saturates* at a far lower rank than the residual, so the redundancy discarded by compression is exactly the high-derivative content the residual amplifies and the field norm cannot see (the decoupling of Sec. IV C). At the shipped ranks the models shrink $51\times$ and $30\times$ at no cost to the certified output.

A. Certified parameter targeting

The first workflow is parameter control [5]: find the free data that realize a chosen physical configuration. The standard tool is a black-box control loop, i.e. an outer Newton/Broyden iteration that adjusts the free data and performs a full constraint solve at every step to measure the resulting physical parameters. We target the ADM mass and angular momentum (M_{ADM}, J) by adjusting (b, q) , over 100 random known-answer targets, and compare three strategies. A *cold* black box solves the constraints from scratch at each control evaluation. A *warm* black box seeds each solve from the interpolant, so every constraint solve begins close to its answer. The

differentiable method runs the entire Gauss–Newton control loop on the free surrogate using the analytic $\partial F / \partial \theta$, then certifies the result with a few Newton steps (i.e. no elliptic solve per step, and no finite-difference Jacobian).

Differentiability changes the cost class here (Fig. 6). The gradient method reaches every target in a *fixed* three certified solves, independent of target difficulty, whereas both black-box variants take a target-dependent median of ten. Warm-starting cuts the per-solve cost but *not* the number of solves: the outer control iteration is unchanged, so a warm start alone never leaves the black box’s complexity class. The distinction is structural. A black-box loop can drive the residual to zero only through function evaluations, and each one is a certified elliptic solve; a low-rank Jacobian update (Broyden’s method,

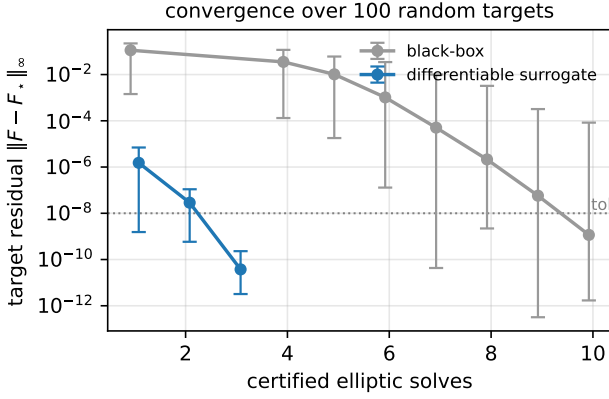


FIG. 6. **Certified parameter targeting.** Reaching a physical target (M_{ADM}, J) by adjusting (b, q) , over 100 random known-answer targets, measured in *certified* elliptic solves. Each method is drawn as the across-target median residual at each solve count, with min-max whiskers. Running the control loop on the free surrogate with the parameter gradient reaches tolerance in at most three certified solves for every target; the black-box loop takes a target-dependent median of ten.

which sidesteps the d extra solves a finite-difference Jacobian would cost) lowers the iteration count but cannot remove the one solve per iteration. The analytic gradient removes those per-iteration solves outright, running the search entirely on the free surrogate and leaving only the last-mile certification, a saving that widens with the number of controlled parameters.

B. Eccentricity reduction

The second workflow is eccentricity control, the most common tuning step in preparing quasi-circular data. We use the Cook effective-potential method [33]: along a sequence of fixed angular momentum J (radial momentum $P_r = 0$, i.e. an apsis), the binding energy $E_b = M_{\text{ADM}} - (M_A + M_B)$ has a minimum at the circular orbit $\partial E_b / \partial b|_J = 0$, and a mildly off-circular apsis at (b_0, J) has eccentricity $e = |b_0 - b'| / (b_0 + b')$ set by the second turning point b' across that minimum. Reducing eccentricity is driving $b_0 \rightarrow b_{\text{circ}}$.

The classical method locates b_{circ} by *scanning* b at fixed J , i.e. one certified solve per point, and fitting the minimum. This workflow needs the tangential momentum P_t as a *free* axis (one scans b at fixed $J = 2bP_t$), which the quasi-circular model of Sec. V deliberately does not carry; we therefore build a dedicated small two-dimensional surrogate in (b, P_t) (equal mass, no spin) by the identical machinery of Secs. IV–V. On it the differentiable binding energy exposes $\partial E_b / \partial b|_J$ analytically, so a Newton root-find locates the circular orbit on the free interpolant and certifies at the end. Sweeping J traces the circular-orbit sequence, whose minimum shifts outward with angular

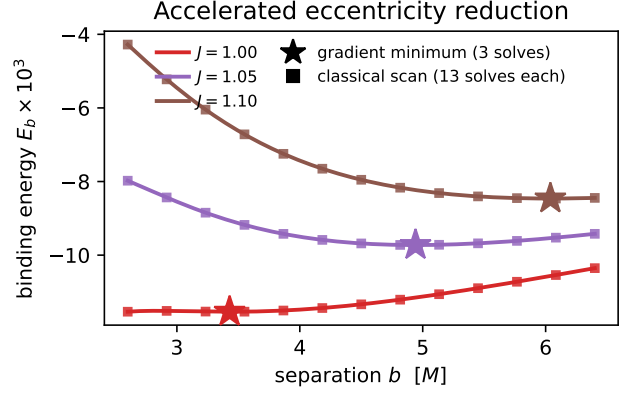


FIG. 7. **Accelerated eccentricity reduction.** At fixed angular momentum J the binding energy $E_b(b)$ has a minimum at the circular orbit; sweeping J the minimum shifts outward, tracing the circular-orbit sequence. For each J the differentiable $\partial E_b / \partial b$ locates that minimum in three certified solves (stars), against the classical certified-solve scan's thirteen (squares). In the shallow small- J well the fixed scan mislocates the minimum, whereas the derivative still resolves it.

momentum (Fig. 7). The gradient locates each b_{circ} in three certified solves against the scan's thirteen, agreeing with it to 5×10^{-3} where the well is well resolved. It is moreover *more accurate* in the shallow, near-innermost-stable-orbit regime, where the fixed thirteen-point scan mislocates the minimum by half a mass and the derivative still resolves it: a black-box scan must sample ever more finely, at more certified solves, exactly where the physics is most delicate. The same differentiable E_b reads the orbital eccentricity off the turning points, from $e = 0$ at the circular orbit to $e \approx 0.23$ at the box edge, and so exposes $\partial e / \partial \theta$ for gradient-based eccentricity reduction.

Both workflows make the same point: the differentiable, certified map turns a black-box search (a control loop and an effective-potential scan) into a gradient problem solved on the free surrogate and certified in a handful of Newton steps, where the black box instead pays a full certified elliptic solve at every trial.

VII. DISCUSSION AND CONCLUSIONS

We have introduced a parametric solver for binary-black-hole puncture initial data that combines sparse interpolation with Newton refinement. The interpolation supplies an inexpensive parameter-dependent initial guess, while the subsequent elliptic solve certifies every returned datum to a prescribed constraint tolerance. Because the construction is implemented in JAX, the resulting parameter-to-solution map is also exactly differentiable with respect to the physical parameters. The four-dimensional aligned-spin and eight-dimensional general-spin models demonstrate that this construction

remains practical across physically relevant binary-black-hole parameter spaces, while the parameter-targeting and eccentricity-reduction examples show how its derivatives can reduce the number of certified elliptic solves required in downstream workflows.

The two headline models are genuinely three-dimensional, quasi-circular parametric surrogates spanning the aligned- and general-spin families (Sec. V). What remains is scope, not a barrier of principle. The most natural extension is *eccentricity*: relaxing the quasi-circularity condition to carry an independent radial momentum adds one axis on top of the orbiting family, by the same machinery — indeed we already build a small (b, P_t) eccentricity model this way (Sec. VIB). The parametric, certified, and differentiable machinery is *dimension-agnostic*, i.e. it sees only a `solve( $\theta$ , guess)` routine and an optional tangent, so each extension is a change of solver and box, not of method.

One limitation is inherited from the free data themselves, not from the parametric method. Bowen–York, conformally flat data carry spurious (“junk”) radiation that the early evolution must shed [34], and cannot represent near-extremal black holes: the *horizon* spin of a Bowen–York puncture saturates at $\chi \approx 0.93$ no matter how large the free-data spin parameter [32, 35], which is why nearly-extremal-spin simulations use conformally curved (e.g. superposed Kerr–Schild) initial data [36]. Within the moving-puncture mainstream ($\chi \lesssim 0.9$) the family built here is the standard input; and because the parametric layer is solver-agnostic (Sec. IV F), a conformally-curved solver could be swapped in under the same interpolation, certification, and compression machinery.

Two accuracy caveats are worth stating explicitly. First, the spin-wall analysis is quantitative only within the physical range ($\chi_{\max} \lesssim 0.8$): pushing the sampling interval toward and past extremal spin mixes the genuine (complex, far) singularity with pre-asymptotic and spatial-floor effects, so the clean single- χ_* characterization is reported for the physical regime, not extrapolated beyond it. Second, the certified $\|R\|_\infty \leq 10^{-10}$ gate is verified at the production grid ($44 \times 32 \times 8$); at higher meridian resolution the whole-field equilibrated residual floors higher, because the stiff m^2/ρ^2 term amplifies roundoff in the azimuthal modes that carry no source content, while the physically populated modes stay three orders of magnitude below. The floor is therefore set by empty modes rather than by the physics, and the certification tolerance should be read at the grid it is quoted for.

Finally, beyond the immediate use cases, the method produces exactly the kind of object a broader surrogate or foundation-model programme for binary-black-hole spacetimes needs at its base: a fast, *exact* (constraint-certified), and *differentiable* initial-data generator across a physical parameter space. Such a generator can serve as a certified training source and a warm-start oracle for those models, supplying gradients $\partial \text{ID} / \partial \theta$ for end-to-end

parameter inference.

ACKNOWLEDGMENTS

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Appendix A: Validation against TwoPunctures

A surrogate is only as trustworthy as the data it reproduces, so we validate the underlying initial data against the established TwoPunctures code [23], specifically its standalone C port from the NRPy suite, compiled against GSL. The comparison is genuine: TwoPunctures discretizes with a Chebyshev² × Fourier grid, LM-initial-data with the prolate single patch of Sec. III A, so agreement is a statement about the physical solution, not a shared discretization. We lead with the most demanding, genuinely non-axisymmetric configurations the quasi-circular model relies on (Appendix A 1), then the quasi-circular data itself (Appendix A 2) and its evolution-readiness (Appendix A 3); the axisymmetric limit, where both codes reduce to their $m = 0$ core and the agreement deepens to machine precision.

1. Non-axisymmetric validation

The quasi-circular model of Sec. V is genuinely non-axisymmetric, so its credibility rests on validating the three-dimensional (Fourier-in- ϕ) solver of Sec. III A against TwoPunctures. We do so on non-axisymmetric configurations that isolate the ingredients the orbiting data combine — an off-axis (transverse) momentum component and a misaligned spin — and on the genuinely three-dimensional observable, the ADM angular momentum, on which the quasi-circular family relies.

For a misaligned-spin slice ($|S^A| = 0.3$ tilted 56.3° off the collision axis, i.e. along $(0.3, 0, 0.2)$; axial momentum $P = 0.5$, $b = 1.5$), the conformal factor converges spectrally to TwoPunctures, $\max |\psi - \psi_{\text{TP}}|$ falling by an order of magnitude over the resolution ladder to 1.7×10^{-7} and still decreasing at the top of the ladder (Fig. 8); this $\sim 10^{-7}$ floor is resolution- rather than solver-limited (one order above the two-centre axisymmetric floor), and the certified polish of Sec. IV C recovers $\|R\|_\infty \leq 10^{-10}$ regardless. The azimuthal content decays exponentially in m (Fig. 8b): a misaligned spin populates $|\hat{u}_1|/|\hat{u}_0| \sim 2\text{--}5\%$ with ~ 2 decades of decay by $m = 3$, while *aligned* spin leaves every $m \geq 1$ mode at machine zero — a hard numerical confirmation that aligned spin is axisymmetric and that a handful of Fourier modes resolve a minimal

break. Across a $b \times |S| \times \theta_S$ grid of 45 slices the certified Newton–Krylov residual floor is uniform and small (median 6.6×10^{-12} , worst 1.8×10^{-11}), never degrading systematically with spin tilt or separation.

While the pointwise field floors at $\sim 10^{-7}$ in this hardest (misaligned-spin) case, the *integral* quantities agree to machine precision even in three dimensions. The total ADM mass matches TwoPunctures to $\sim 10^{-12}$ on off-axis-momentum data (where the mode iteration is the full Newton). The genuinely three-dimensional observable, the ADM angular momentum, is computed from the conformal Bowen–York tensor by a York surface integral and cross-checked against the closed form $J^i = \sum_X ((S^X)^i + (x_X \times P_X)^i)$ and against TwoPunctures (Fig. 9): for pure spin the ADM- J tilt off the collision axis equals the spin tilt to machine precision across all magnitudes and separations; an anti-symmetric off-axis momentum P_x produces an orbital $J_y = 2bP_x$ recovered exactly by the surface integral; and agreement with the TwoPunctures-reported J (rotated to the common frame) is at the level of the analytic Gauss law on both sides, $\leq 1.4 \times 10^{-14}$. In the axisymmetric limit the field agreement itself deepens to machine precision — two independently discretized spectral codes agreeing on ψ to 4.7×10^{-12} and on the total ADM mass to a relative 1.0×10^{-11} (equal-mass slice $b = 3$, axial $P = 0.5$) — which we quote as the deepest code-to-code anchor rather than as a separate axisymmetric study.

2. Quasi-circular data against TwoPunctures

The headline check is that the quasi-circular initial data itself reproduces the oracle. The frame convention (risk R2) is fixed first: the separation is along z and the quasi-circular tangential momentum along x , so the orbital angular momentum is along $+y$; the proper rotation taking the LM-initial-data frame to TwoPunctures maps our y to TP- z , and “aligned spin” therefore means the S_y component. With the convention fixed, the momentum condition is honest: the closed-form tangential momentum approaches the Newtonian value, $P_t/[\mu\sqrt{M/2b}] \rightarrow 1$ as b grows (from 1.59 at $b = 2$ to 1.016 at $b = 64$), and the radial component vanishes, $P_r/P_t \rightarrow 0$ (as $b^{-5/2}$), so “quasi-circular” is quantitatively justified. The field and mass converge to the oracle (Fig. 9b,c): on a quasi-circular slice ($b = 4$, $q = 1$, no spin) the conformal factor converges spectrally to TwoPunctures, $\max|\psi - \psi_{\text{TP}}|$ reaching 1.7×10^{-9} at 64×44 (the two-centre spatial floor), and the total ADM mass agrees with the TwoPunctures energy to 5.8×10^{-12} — the boundary-spectral M_{ADM} converging far deeper than the pointwise field. Finally, the closed-form angular momentum $J = 2bP_t + S_y^A + S_y^B$ (along $+y$) equals the frame-rotated TwoPunctures ADM J to machine precision, with net linear momentum $P_A + P_B = 0$ exactly; since the Bowen–York J is analytic in the free data on both sides, this pins the frame convention and momen-

tum wiring (retiring risk R2) rather than measuring the field.

3. Evolution-readiness on an evolution grid

The oracle-independent test of evolution-readiness is to interpolate the spectral initial data onto a uniform Cartesian grid — as an evolution code would ingest it — and measure the residual constraint violation with a *generic* finite-difference Hamiltonian and momentum constraint monitor (FD Christoffels and Ricci, no conformal-flatness shortcut), exactly as a free-evolution code’s diagnostics would [37]. On the equal-mass head-on base case both constraints converge at the expected second order in the grid spacing h (measured orders 2.14 for $\|H\|_2$ and 2.02 for $\|M\|_2$). Critically, feeding the *TwoPunctures*-sourced ψ through the identical monitor on the same grid gives a violation identical to all recorded digits (the two ψ fields differ far below the truncation-dominated norms) — so the residual is genuine finite-difference truncation, not a defect of the LM-initial-data solution, and our initial data is as evolution-ready as TwoPunctures’. Equivalently, the spatial resolution used here is *not* the accuracy-limiting factor for a numerical-relativity evolution: the initial-data spectral error lies well below the finite-difference truncation of any practical evolution grid, so the elliptic solve is already resolved beyond what the downstream evolution can register, and the certified polish guarantees $\|R\|_\infty \leq 10^{-10}$ on the emitted datum in any case.

Appendix B: Verification of the parameter sensitivities

The verification proceeds in two stages, because two distinct statements are at issue: that Eq. (23) is the sensitivity of the certified solve, and that the gradient the surrogate exposes is that same sensitivity.

Table I settles the first on the production three-dimensional operator, comparing the tangent of Eq. (23) against central finite differences of the certified Newton–Krylov solve itself. The two agree to $\sim 10^{-9}$ on every axis of the aligned-spin box, at the accuracy of the finite-difference oracle, so the analytic $\partial R/\partial\theta_k$ and the back-solve reproduce the true derivative of the certified map.

Table II settles the second. Automatic differentiation of the surrogate agrees with central finite differences of that same surrogate to 10^{-10} – 10^{-9} , which verifies the differentiation itself but says nothing about certification. The load-bearing column is the last one: the exposed gradient reproduces the independently computed tangent of Eq. (23) to $\sim 2 \times 10^{-7}$ along the spin axes and $\sim 10^{-5}$ along b and q . That residual is the interpolation error of the derivative, not an error in either object, and it is largest along b and q for the same reason the field

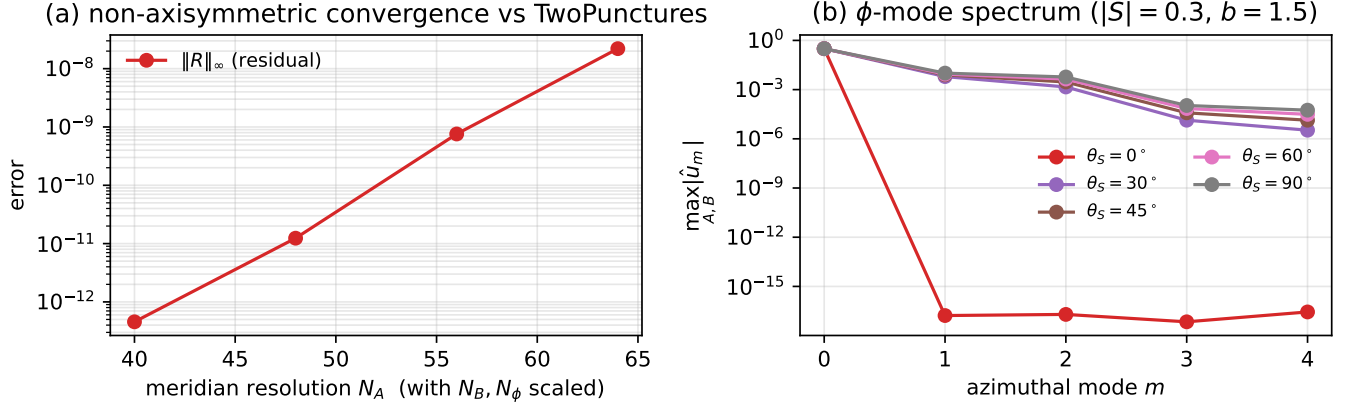


FIG. 8. **Non-axisymmetric data validated against TwoPunctures.** *Left:* for a misaligned-spin slice ($\mathbf{S}^A = (0.3, 0, 0.2)$, $b = 1.5$, $P = 0.5$), the conformal factor converges spectrally to TwoPunctures ($\max |\psi - \psi_{\text{TP}}|$, dashed), still decreasing at the top of the resolution ladder, while the certified Newton–Krylov residual $\|R\|_\infty$ (solid) is near machine precision at the production resolution. Its slow rise at higher meridian resolution is the empty-high- m -mode roundoff amplification of Sec. VII, a norm artifact rather than a solver limitation: the field keeps converging throughout. *Right:* the azimuthal ϕ -mode spectrum of the solved field decays exponentially in m , and aligned spin ($\theta_S = 0$) leaves every $m \geq 1$ mode at machine zero, confirming that few Fourier modes resolve a minimal break of axisymmetry.

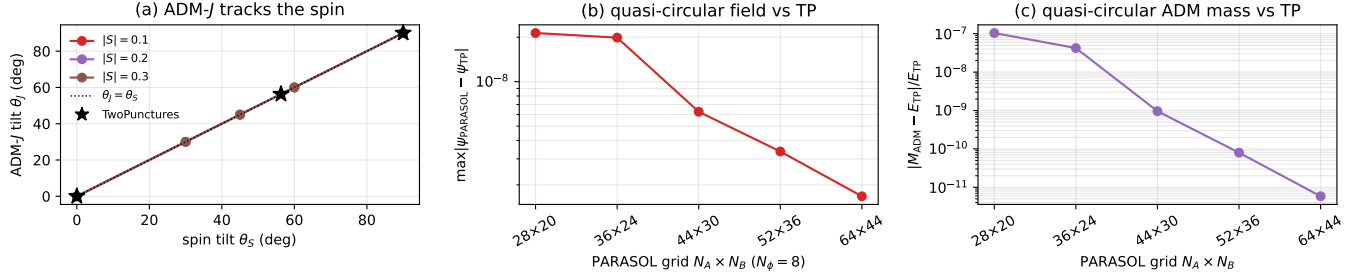


FIG. 9. **TwoPunctures validation: the three-dimensional observable and the quasi-circular data.** (a) For pure spin the ADM- J tilt off the collision axis tracks the spin tilt exactly ($\mathbf{J} = \sum_X \mathbf{S}^X$): all magnitudes collapse onto $\theta_J = \theta_S$, with the TwoPunctures anchors (stars) on the line. (b,c) On a quasi-circular slice ($b = 4$, $q = 1$, no spin) the conformal factor converges spectrally to TwoPunctures, saturating at the two-centre spatial floor, while the ADM mass converges some three orders of magnitude deeper still — the boundary-spectral M_{ADM} outrunning the pointwise field.

interpolation is slowest there (Sec. VB); it decreases ex-

ponentially with the parameter order.

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TABLE I. **The tangent of Eq. (23) against the certified solve.** Maximum relative difference between the implicit-function-theorem tangent $\partial U/\partial \theta_k$, obtained from one back-solve against the factored Jacobian, and second-order central finite differences of the certified Newton–Krylov solve with step h . Non-axisymmetric quasi-circular slice at $b = 2.4$, $q = 1.7$, $P_t = 0.5$, $\chi_y^A = 0.30$, $\chi_y^B = 0.20$, on the $(N_A, N_B, N_\phi) = (18, 14, 6)$ grid; both solves are certified to $\|R\|_\infty = 2.4 \times 10^{-14}$.

Axis θ_k	h	Eq. (23) vs. finite differences
b	10^{-4}	1.9×10^{-9}
q	10^{-4}	2.7×10^{-9}
χ_y^A	10^{-4}	7.3×10^{-9}
χ_y^B	10^{-4}	5.6×10^{-9}

TABLE II. **The exposed sensitivity against both references.** Maximum relative difference between the automatic-differentiation sensitivity $\partial U/\partial \theta_k$ of the surrogate and, respectively, second-order central finite differences of the same surrogate with step h , and the implicit-function-theorem tangent of Eq. (23) computed independently from the certified solve at the same parameter point. Two-centre aligned-spin interpolants on the $(N_A, N_B) = (36, 24)$ grid at $M = 1$: a (b, q) model of order $Q = (12, 10)$ over $b \in [3, 12]$, $q \in [1, 3]$, evaluated off-node at $(b, q) = (6.337, 1.713)$; and a (χ_y^A, χ_y^B) model of order $Q = 8$ over $[0, 0.6]^2$ at fixed $q = 1.5$, $b = 4$, evaluated off-node at $(\chi_y^A, \chi_y^B) = (0.413, 0.227)$, where χ_y^A and χ_y^B are the aligned spin components on the two punctures. The reference solves are certified to $\|R\|_\infty \leq 2 \times 10^{-12}$.

Axis θ_k	h	vs. finite differences	vs. Eq. (23)
b	10^{-6}	5.5×10^{-9}	1.3×10^{-5}
q	10^{-6}	4.4×10^{-10}	1.2×10^{-5}
χ_y^A	10^{-6}	2.0×10^{-10}	1.7×10^{-7}
χ_y^B	10^{-6}	2.0×10^{-10}	1.8×10^{-7}

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